MILITARY SPECIFICATION

FUZE, MECHANICAL TIME, M591, ASSEMBLY

1. SCOPE

1.1 This specification covers Fuze, Mechanical Time, M591 Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-10464	- Cans, Hermetic Sealing, Metal Light Gage Tear-strip Type
MIL-B-60856	- Booster, M125A1, Loading and Packing

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-331	- Fuze and Fuze Components Environmental and Performance Test for
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions

F10535728	- Fuze, Mechanical Time, M591, Assembly
F8595541	- Booster, M125A1, Assembly
C8794405	- Box, Packing, Ammunition for Fuze

FSC 1390

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2.1 (Cont'd)

F8794406	- Ammunition Container, Metal, for Fuze
F10535730	- Arrangement of Timers for Testing M591 Fuze
IEL-10535728	- Inspection Equipment List for Fuze, MT, M591 Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards, and specifications, except as provided in 6.1.

3.2 Dynamic regulation.-The movement subassembly shall be regulated to the frequency specified on the applicable drawing. The timing error between 0 to 95 seconds shall not exceed 0.07 percent while spinning at the specified revolutions per minute (rpm).

3.3 Environmental.

3.3.1 Jolt.-The fuze shall comply with MIL-STD-331, Test No. 101.1, except that the fuze shall be tested less the booster pellet and lead cup.

3.3.2 Jumble.-The fuze shall comply with MIL-STD-331, test NO. 102.1, except that the fuze shall be tested less the booster pellet and lead cup.

3.3.3 Forty-foot drop (in metal container).-The fuze shall comply with MIL-STD-331, Test No. 103, except that the fuze shall be tested less the booster pellet and lead cup.

3.3.4 Transportation-Vibration.-The fuze shall comply with MIL-STD-331, Test No. 104, except that the fuze shall be tested less the booster pellet and lead cup.

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3.4 Booster assembly.-The booster assembly shall comply with all the requirements of Drawing F8595541 and MIL-B-60856.

3.5 Fuze assembly.

3.5.1 Function.-Timer No. 2 and Timer No. 3 assemblies shall function within the setting range of ten (10) to one hundred (100) seconds and within the temperature range of minus 40 degrees to plus 160 degrees Fahrenheit (F).

3.5.2 Safety.-The fuze assembly shall be incapable of functioning from zero (0) to 7.0 seconds. The fuze shall not function prematurely (see 6.3).

3.6 Ballistic.

3.6.1 Time-to-Burst (Ambient temperature).-The fuze assembly as modified for testing in accordance with the provisions of 4.4.13 shall initiate detonation of the test round within the time limits specified in Table I, when fired as follows, (for definition of test fuze, see 6.3.4).

TABLE I

Weapon	Charge	Timer Setting	Upper-Limit	Lower Limit
8 inch How	Zone 1	20 seconds	21.029	19.799
8 inch How	Zone 7	55 seconds	56.216	54.366

3.6.2 Low temperature.-The fuze assembly as modified for testing in accordance with the provisions of 4.4.13 shall initiate detonation of the test round when the temperature of the fuze, projectile and propellant is maintained at minus 40 degrees, minus 5 degrees Fahrenheit and fired as follows:

Weapon	Charge	Timer Setting
8 inch How	Zone 1	20 seconds

3.6.3 High temperature.-The fuze assembly as modified for testing in accordance with the provisions of 4.4.13 shall initiate detonation of the test round when the temperature of the fuze, projectile and propellant is maintained at plus 160 degrees, plus 5 degrees Fahrenheit and fired as follows:

Weapon	Charge	Timer Setting
8 inch How	Zone 7	55 seconds

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3.7 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.-All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.7.2 Surface finish.-The requirements for surface finish are as detailed on the applicable drawings.

3.7.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing.

3.7.4 Burr.-No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.5 Foreign matter.-No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter. The methods and agents used in processing and cleaning shall not adversely affect the finish or contaminate the parts.

3.7.6 Protective finish.-Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Lot formation.-Each inspection lot of fuzes shall be comprised of components (except explosive components) each produced by one manufacturer by one process in accordance with the same drawing and revision and specifications and revisions, as applicable, and assembled on one assembly line. Each explosive component within a lot of assembled fuzes shall be limited to one manufacturer and one interfix number.

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4.2 First article sample.-When required, the first article sample shall be as specified in the contract and shall consist of the following:

200 (aggregate) component parts as delineated in 4.2.2

625 (aggregate) subassemblies as delineated in 4.2.3

36 (aggregate) fuze assemblies as delineated in 4.2.5 and 4.2.6

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1 Examination and test.-After inspection by the contractor and provisional acceptance at source by the Government, 24 fuze assemblies shall be forwarded to a designated proving ground as specified in the contract. Two hundred (200) component parts, 625 subassemblies and 12 fuze assemblies shall be inspected, as provided herein, at a Government laboratory or such other facility as specified in the contract.

4.2.2 Components.

4.2.2.1 Twenty (20) of each of the (9) components shall be tested as in 4.3.3.2 (Hardness), 4.3.3.3 (Lead), and 4.3.3.5(Bend).

4.2.2.2 Twenty (20) of each component (1) shall be tested as in 4.3.3.4 (Compression).

4.2.3 Subassemblies.

4.2.3.1 Twenty-five (25) of each of the (21) subassemblies shall be tested as in 4.3.3.6.1 (Static Load).

4.2.3.2 Twenty-five (25) of each of the (4) subassemblies shall be tested as in 4.3.3.6.2 (Torque).

4.2.4 Acceptance criteria.-Failure of any component or subassembly to comply with the requirement of the applicable drawing for the pertinent characteristics shall be cause for rejection of the first article sample.

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4.2.5 Environmental tests.-Twelve (12) fuze assemblies shall be subjected to the following MIL-STD-331 tests with the fuze timers set on safe (s).

Test	Quantity	Requirement Para.	Test Method
Jolt	3	3.3.1	4.4.12.1
Jumble	3	3.3.2	4.4.12.2
Forty Foot Drop	3	3.3.3	4.4.12.3
Transportation-Vibration	3	3.3.4	4.4.12.4

4.2.5.1 Acceptance criteria.-The criteria of the applicable MIL-STD shall apply as determined by disassembly, examination and tests in accordance with the procedure in 4.4.12. Where the accept/reject decision is based on engineering judgement, concurrence by the responsible technical agency must be obtained.

4.2.6 Ballistics.-Twenty-four (24) fuze assemblies of the first article sample shall be prepared as specified in 4.4.13 and subjected to the following ballistics tests, (for definition of test fuze, see 6.3.4).

Phase	Zone	Quantity of Test Fuzes	Test Fuze Setting	Requirement	Test Method
Time-to-burst (ambient)	1	18	20	3.6.1	4.4.13
Time-to-burst (ambient)	7	18	55	3.6.1	4.4.13
Low Temperature	1	6	20	3.6.2	4.4.13
High Temperature	7	6	55	3.6.3	4.4.13

4.2.6.1 Acceptance criteria.-The occurrence of any of the following shall be cause for rejection of the first article sample:

a. A premature function in any phase (see 6.3.1).
b. More than two duds in either high or low temperature phase (see 6.3.3).
c. One or more unsettable timer in the high temperature phase.
d. More than one unsettable timer in the low temperature phase.
e. Any of the following in the ambient time-to-burst phases:

(1) One or more unsettable timers.
(2) More than one dud and one outlier-offtime or two outlier-offtime in the combined phases.
(3) Failure to meet the limits specified in Table I by an estimated lot percent defective greater than the appropriate value given in Table II (see 4.4.13.1.1).

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4.2.6.1 (Cont'd)

TABLE II

Zone 1 Set 20 Sec.

Sample Size	Allowable % Defective
18	2.07
17	2.08
16	2.09
15	2.11

Zone 7 Set 55 Sec.

Sample Size	Allowable % Defective
18	2.07
17	2.08
16	2.09
15	2.11

4.3 Inspection provisions.

4.3.1 Submission of product.-The product shall be presented in accordance with MIL-STD-105 (except where specified otherwise).

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major defects shall be performed on a single defect basis in accordance with MIL-STD-105. For each classification of defects contained herein, the AQL given shall be applied to each defect contained herein. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.3.2.1 Housing, Middle and Lower-Drawings F8594382 and F8594460 covering a detail of Drawing F10535728.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR	AQL = 1.0%	
101.	Outside diameter exceeds max. allowable	Gage
102.	Location of zero line, incorrect	Gage
103.	Marking missing, incorrect or illegible	Visual
104.	Protective finish, improper	Visual

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4.3.2.2 Dial, Middle and Lower Drawings F8594385 and F8594386 covering a detail of Drawing F10535728.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR	AQL = 1.0%	
101.	Cumulative error of time scale exceeds max. (to be measured from zero line to 20, 50, 80 second lines)	Gage
102.	Location of safety line S, incorrect	Gage
103.	Location of numerals, improper	Visual
104.	Marking missing, incorrect or illegible	Visual

4.3.2.3 Block, Breech, Drawing F8594565 covering a detail of Drawing F10535728.

CRITICAL		
	None defined	
MAJOR	AQL = 1.0%	
101.	Major diameter of large external thread exceeds max. allowable	Gage
102.	Pitch diameter of large external thread incorrect	Gage
103.	Major diameter of small external thread exceeds max. allowable	Gage
104.	Pitch diameter of small external thread incorrect	Gage
105.	Thread damaged or not full	Visual
106.	Flash holes (2) not broken through	Visual

4.3.2.4 Pin, Firing Drawing C8594434 covering a detail of Drawing F10535728.

CRITICAL		
	None defined	

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 1.0%	
101. Diameter of striker point, incorrect	Gage
102. Surface finish on shoulder angle, improper	Visual

4.3.2.5 Firing Arm Assembly, Drawing C8593094 covering a detail of Drawing F10535728.

CRITICAL	
1. Firing arm upright fractured, broken or deformed	Visual
2. Firing arm not staked to shaft	Visual
MAJOR AQL = 0.40%	
101. Insecure weight	Manual/Visual
102. Metal burred or damaged	Visual

4.3.2.6 Rotor Assembly, Drawing D8594410, covering a detail of Drawing F10535728.

CRITICAL	
1. Primer above flush of rotor surface	Visual
MAJOR AQL = 0.40%	
101. Crimping inadequate	Visual
102. Primer installed incorrectly (red end facing cushion)	Visual

4.3.2.7 Delay Arming Mechanism Assembly, Drawing F8594387, covering a detail of Drawing F10535728 (to be checked just prior to assembly into housing).

CRITICAL	
1. Rotor not in the unarmed position (to be checked just prior to assembly to the movement - see 6.2.2)	Visual
2. Escapement lever pallet(s) missing, damaged or deformed	Visual

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4.3.2.7 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 0.40%	
101. Assembly not fully wound (centrifugal gear not against stop as viewed through the elongated hole - to be checked prior to each spin test)	Visual
102. Escapement spring broken, deformed or unseated	Visual
103. End of "D" shaft spring protrudes beyond diameter	Visual
104. Escapement lever not locked on proper side of pallet lock slides	Visual
105. Centrifugal gear springs project into body assembly screw holes	Visual
106. Pins or protrusions above flush of either surface	Visual
107. Foreign matter	Visual

4.3.2.8 Movement Subassembly (after dynamic regulation, except as otherwise specified) Drawing F8593032, covering a detail of Drawing F10535728.

CRITICAL	
1. Firing pin fires on 'not fire' position of firing arm	Gage
2. Extension of firing arm upright above top plate less than minimum allowable (with firing arm shaft shouldered on plate No. 2)	Gage
MAJOR AQL = 0.40%	
101. Firing pin does not fire on "must fire" position	Gage
102. Firing pin protrusion less than min. in fired position	Gage
103. Extension of firing arm upright above top plate exceeds maximum (with firing arm shaft shouldered against top plate)	Gage
104. Firing arm shaft spring protrudes	Visual
105. Burrs and Irregularities which prevent proper seating of movement on the delay arming mechanism	Visual
106. Firing pin striker point damaged	Visual
107. Foreign matter	Visual
108. Staking of movement assembly screws inadequate (check prior to regulation)	Visual

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4.3.2.9 Timing Disc Assembly, Drawing C8594472, covering a detail of Drawing F10535728.

Defect	Method of Inspection
CRITICAL	
1. Safety disc tabs not engaged in timing disc bushing slots	Visual
MAJOR AQL = 0.40%	
101. Adjusting segment extends beyond periphery of timing disc	Visual
102. Safety disc does not extend beyond periphery of timing disc	Visual
103. Improper or incomplete assembly	Visual

4.3.2.10 Movement Assembly, Drawing F8594465 covering a detail of Drawing F10535728.

CRITICAL	
1. Timing disc bushing loose on knurl of arbor (check prior to assembly of locknut)	Manual
2. Timing disc retainer insecure, damaged or missing	Manual/Visual
3. Distance from top of top plate to top of retainer exceeds maximum allowable (after staking of retainer)	Gage
4. Firing pin in fired position	Visual
MAJOR AQL = 0.40%	
101. Total height exceeds max. allowable	Gage
102. Clearance between plate No. 7 and safety disc less than minimum allowable	Gage
103. Clearance between adjusting segment and underside of retainer less than minimum allowable	Gage
104. Escapement lever not locked on proper side of pallet lock slides (F8593032)	Visual
105. Escapement spring broken, deformed or unseated	Visual
106. Pins or protrusions above flush of bottom surface	Visual

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4.3.2.11 Timers, Numbers 2 and 3 Subassemblies, Drawings D8594471 and D8594463, covering details of Drawing F10535728.

Defect	Method of Inspection
CRITICAL	
1. Edge of firing arm outside of safety disc setting lines	Visual
MAJOR AQL = 0.40%	
101. Location of arbor shaft	Gage
102. Height of arbor above surface "A" exceeds max.	Gage
103. Insecure assembly	Manual
104. Foreign matter	Visual
105. Burr	Visual

4.3.2.12 Dial Assemblies, Drawings D8594468 and D8594469, covering a detail of Drawing F10535728.

CRITICAL	
None defined	
MAJOR AQL = 0.40%	
101. Setting pin protrusion beyond surface A, incorrect	Gage
102. Staking of setting pin inadequate	Visual
103. Hammerspring rubs against setting pin	Visual
104. Metal burred or damaged	Visual

4.3.2.13 Timer No. 2 and 3 Assemblies, Drawings F8594397 and F10523941 (before assembly of dial plug), covering details of Drawing F10535728.

CRITICAL	
1. Setting pin engagement with timing disc lug less than minimum allowable with end play removed, i.e., arbor down, (check after lockwire tensioning operation)	Gage
2. Dial assembly not seated properly in housing	Visual

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4.3.2.13 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 0.40%	
101. Clearance between firing arm upright and setting pin less than min. allowable	Gage
102. Pellet holder location with respect to bottom surface of timer No. 3 housing, incorrect (Drawing F10523941)	Gage
103. Insecure assembly	Manual
104. Incorrect zero setting	Gage
105. Hammerspring not over lug when timer is set on Safe(s)	Visual
106. Pellet holder assembly not in correct hole in timer No. 3 Assembly	Visual

4.3.2.14 Fuze Subassembly, Drawing F10551148 covering a detail of Drawing F10535728.

CRITICAL

None defined

MAJOR AQL = 0.40%	
101. Total length exceeds maximum allowable	Gage
102. Eccentricity of housing(s) with respect to large external thread	Gage
103. Insecure assembly	Manual
104. Gap between housings or at breech block interface	Visual
105. Marking missing, incorrect or illegible	Visual
106. Thread damaged	Visual

4.3.2.15 Booster Assembly, Drawing F8595541, covering a detail of Drawing F10535728.

CRITICAL

1. Rotor not fully engaged by both spring loaded detents (to be checked just prior to assembly to the fuze)	Visual

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4.3.2.15 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 0.40%	
101. Inadequate assembly; loose or missing parts	Manual/Visual
102. Staking (of rotor cover, closing cup, pins) inadequate	Visual
103. Foreign matter	Visual
104. Thread damaged	Visual

4.3.2.16 Fuze, MT, M591 Assembly, Drawing F10535728.

CRITICAL	
1. Timer not set at safe(s) position	Visual
MAJOR AQL = 0.40%	
101. Overall length exceeds maximum allowable	Gage
102. External body thread damaged	Visual
103. Staking (of booster to fuze) inadequate	Visual
104. Foreign matter	Visual
105. Burr	Visual

4.3.2.17 Ammunition Container, Metal Drawing F8794406.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR AQL = 0.65%	
201. Contents loose	Manual
202. Key missing or damaged	Visual
203. Container punctured, dented, damaged or deformed	Visual
204. Marking missing, incorrect or illegible	Visual

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4.3.2.18 Box, Packing, Drawing B8794405.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
None defined		
MINOR	AQL = 0.65%	
201.	Contents loose	Manual
202.	Hardware missing, broken, loose or improperly engaged	Manual/Visual
203.	Rope handles insecure, damaged or missing	Manual/Visual
204.	Strapping missing, broken, loose or improperly assembled	Manual/Visual
205.	Car seal missing, unsealed or improperly positioned (when applicable)	Visual
206.	Box damaged	Visual
207.	Marking missing, incorrect or illegible	Visual

4.3.2.19 Process control audit.-Periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings. Bending adjustment of firing arm upright is not permitted.

a. Packing Box, Drawing F8794405, and Metal Container, Drawing F8794406.

(1) Packing operations

b. Timer No. 2 and 3 Assemblies, Drawings F8594397 and F10523941.

(1) Zero setting procedure

(2) Insertion of pellet screw assembly in Timer No. 3

(3) Application of sealant

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4.3.2.19 (Cont'd)

c. Delay Arming Mechanism, Drawing F8594387.

(1) Application of lubricant

(2) Assembly screw torquing operation

(3) Cocking and winding operation prior to each spin test

d. Plate No. 1 and 2 Assembly, Drawing D8594404.

(1) Application of lubricant

e. Movement Assembly, Drawing F8594465.

(1) Locknut torquing operation prior to timing disc torque test

f. Movement Subassembly Drawing F8593032.

(1) Staking of top assembly screw and torquing of bottom assembly screws prior to dynamic regulation.

(2) Application of lubricant

(3) Winding of mainspring before assembly of screw

(4) Dynamic regulation operation

(5) Winding of movement after dynamic regulation

(6) Gear train backlash removal

g. Laminated Plate No. 1 and No. 2 Assembly, Drawing F8593043.

(1) Application of lubricant

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.-The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable standards or specifications listed on the drawing (see 4.4.1).

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4.3.3.2 Hardness (destructive).-Twenty samples from each lot of the following components shall be subjected to the hardness test. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot. Where plating is required, the sample parts submitted shall be unplated (see 4.4.2).

Component	Drawing Number
Gear, Escapement	D8594412
Arbor, Escapement	D8594416
Lever, Escapement (Movement)	F8593023
Arm, Firing	C8593095
Guide, Adjusting	C8594273
Lever, Escapement (Delay Arming Mechanism)	F8594400
Spring, Hammer	C8594431
Spring, Timing Disc	C8594454
Washer, Timing Disc	B8594267

4.3.3.3 Load-Timing Disc Spring (destructive).-The same 20 springs subjected to the hardness test shall be subjected to the load specified on Drawing C8594454. Failure of any spring to comply with the requirement shall be cause for rejection of the lot (see 4.4.3).

4.3.3.4 Compression-Firing Pin Spring (nondestructive).-Twenty (20) from each lot shall be subjected to the compression test specified on Drawing C8594315. Failure of any spring to comply with the requirement shall be cause for rejection of the lot (see 4.4.4).

4.3.3.5 Firing arm bend (destructive).-The same 20 firing arms subjected to the hardness test shall be subjected to the bend test as specified on Drawing C8593095. Failure of any firing arm to comply with the requirements shall be cause for rejection of the lot (see 4.4.5).

4.3.3.6 Static load and torque.

4.3.3.6.1 Static load.-Twenty-five (25) samples from each lot of the following subassemblies shall be subjected to the static load test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (see 4.4.6 and 6.7).

Subassembly	Drawing Number
Centrifugal Drive Gear Assembly	D8594414
Centrifugal Drive Gear Assembly	D8594473
"D" Shaft Assembly	C8594415
No. 1 Gear and Pinion Assembly (D.A.M.)	C8594418
No. 2 Gear and Pinion Assembly (D.A.M.)	C8594419

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4.3.3.6.1 (Cont'd)

Subassembly	Drawing Number
No. 3 Gear and Pinion Assembly (D.A.M.)	C8594420
Escapement Gear & Pinion Assembly (D.A.M.)	D8594411
Escapement Lever Assembly (D.A.M.)	D8594406
Pallet Lock Assembly (D.A.M.)	B8593152
Rotor Assembly (D.A.M.)	D8594410
Firing Arm Assembly (Pin & Arm)	C8593094
Escapement Gear & Pinion Assembly (Movement)	C8593382
Escapement Lever Assembly (Movement)	F8593015
Pallet Lock Assembly (Movement)	C8593016
Lever Assembly (Movement)	B8593124
No. 1 Gear & Pinion Assembly (Movement)	C8594280
No. 2 Gear & Pinion Assembly (Movement)	C8594283
No. 3 Gear & Pinion Assembly (Movement)	C8594286
No. 4 Gear & Pinion Assembly (Movement)	C8593085
Disc and Stud Assembly (Movement)	C10520614
*Arbor and Housing Assembly (Pin-Nondestructive)	D8594466 (reference only)

* #5 Gear 8593102 and Arbor 8593091 (Destructive)

4.3.3.6.2 Torque.-Twenty-five (25) samples from each lot of the following subassemblies shall be subjected to the torque test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (see 4.4.6 and 6.7).

Subassembly	Drawing Number
Disc and Stud Assembly (Movement)	C10520614
Escapement Lever Assembly (D.A.M.)	D8594406
Escapement Lever Assembly (Movement)	F8593015
*Arbor and Housing Assembly	D8594466 (reference only)

* #5 Gear 8593102 and Arbor 8593091 (destructive)
Gear 8593102 and Arbor 8593091 only shall be assembled for static load and torque test sample using the same assembly methods and equipment that shall be used in production of

4.3.3.7 Middle and lower deal assemblies.

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4.3.3.7.1 Hammerspring deflection (nondestructive).-The hammerspring of each dial assembly produced shall be subjected to the load specified on Drawing D8594468 or D8594469 as applicable. Failure to comply with the requirements shall be cause for rejection of the dial assembly (see 4.4.7.1).

4.3.3.8 Movement subassembly.

4.3.3.8.1 Kinetic energy of firing pin (nondestructive).-Fifty (50) movement subassemblies from each lot shall be tested to determine compliance with the requirement specified on Drawing F8593032. Failure of any assembly to conform to the requirement shall be cause for rejection of the lot of movement subassemblies (see 4.4.8.1).

4.3.3.8.2 Dynamic regulation (nondestructive).-The regulation tape record (95 seconds) of each movement subassembly produced shall be inspected to determine acceptability of the movement. Any movement subassembly which fails to comply with the requirement of 3.2 shall be rejected (see 4.4.8.2).

4.3.3.8.3 Spin function (nondestructive).-The pallet lock and lever assemblies within each movement subassembly produced shall be tested to determine compliance with the function requirement of Drawing F8593032. Failure to comply with the requirement shall be cause for rejection of the subassembly (see 4.4.8.3).

4.3.3.8.4 Spin nonfunction (nondestructive).-Each movement subassembly produced shall be tested to determine compliance with the nonfunction requirement of Drawing F8593032. Failure of the pallet lock slides to remain in the locked position shall be cause for rejection of the subassembly (see 4.4.8.4).

4.3.3.9 Timing disc torque (nondestructive).-The timing disc of each movement assembly produced shall be subjected to the torque specified on Drawing F8594465. Failure to comply with the requirement shall be cause for rejection of the movement assembly (see 4.4.9).

4.3.3.10 Delay arming mechanism.

4.3.3.10.1 Spin arming (nondestructive).-Each cocked and fully wound delay arming mechanism assembly produced shall be tested to determine compliance with the arming and time requirements of Drawing F8594387. Failure to comply with the requirements for each spin phase shall be cause for rejection of assembly (see 4.4.10.1 and 6.2.1).

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4.3.3.10.2 Spin nonarming (nondestructive).-Each cocked and fully wound delay arming mechanism assembly produced shall be tested to determine compliance with the nonarming requirement of Drawing F8594387. Failure of the pallet lockslides to remain in the locked position shall be cause for rejection of the assembly (see 4.4.10.2).

4.3.3.11 Timer No. 2 and No. 3 assemblies.

4.3.3.11.1 Setting torque (nondestructive).-Each timer assembly produced shall be tested to determine compliance with the torque requirement of Drawing F8594397 and F10523941 respectively. Failure to comply with the requirement shall be cause for rejection to the timer assembly (see 4.4.11.1).

4.3.3.12 Environmental tests (Lots - Destructive).-Twelve (12) assemblies shall be selected from each production lot and subjected to the following MIL-STD-331 tests with the fuze timers set on safe (s).

Test	Quantity	Req. Para.	Test Method
Jolt	3	3.3.1	4.4.12.1
Jumble	3	3.3.2	4.4.12.2
Forty Foot Drop (in can)	3	3.3.3	4.4.12.3
Transportation-Vibration	3	3.3.4	4.4.12.4

4.3.3.12.1 Acceptance criteria.-The criteria of the applicable MIL-STD shall apply as determined by disassembly, examination and tests in accordance with the procedures in 4.4.12. Where the accept/reject decision is based on engineering judgment, concurrence by the responsible technical agency must be obtained.

4.3.3.13 Ballistic tests (destructive).-Eighteen fuzes from each production lot shall be prepared as specified in 4.4.13 and subjected to the following ballistics tests (for definition of test fuze, see 6.3.4).

Phase	Zone	Quantity of Test Fuzes	Set Time (sec)	Requirement	Test Method
Time-to-burst (Amb.)	1	18	20	3.6.1	4.4.13
Time-to-burst (Amb.)	7	18	55	3.6.1	4.4.13

4.3.3.13.1 Acceptance criteria.-The occurrence of any of the following shall be cause for rejection of the lot the ballistic sample represents:

a. One or more unsettable timers
b. A premature function in either phase (6.3.1)
c. More than one dud and one outlier-offtime or two outlier-offtime in the combined phases (6.3.3)
d. Failure to meet the limits specified in Table I by an estimated lot percent defective greater than the appropriate value given in Table II (see 4.4.13.1.1).

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4.3.3.14 Container, air pressure test (nondestructive).-Each sealed container shall be tested to determine compliance with the requirement of Drawing F8794406. Failure to comply with the requirement shall be cause for rejection (see 4.4.14).

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed in IEL-10535728. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Protective finish.-The test procedures for protective finish shall be in accordance with the applicable specification referenced on the component drawing.

4.4.2 Hardness.-Samples shall be tested in an area which is solidly supported in order to obtain valid readings. The median value of three (3) hardness readings taken on each sample unit shall be used as the individual value.

4.4.3 Load - timing disc spring.-The test shall be performed using the values as specified on Drawing C8594454. The timing disc springs that have been subjected to the load test shall be scrapped.

4.4.4 Compression.-The firing pin springs shall be compressed for the length of time specified on Drawing B8594315.

4.4.5 Firing arm bend.-The test shall be performed as specified on Drawing C8593095. The firing arms that have been subjected to the bend test shall be scrapped.

4.4.6 Static load and torque.-The test shall be performed using the values as specified on the applicable drawing. All #5 gear and arbor assembly samples tested shall be scrapped.

4.4.7 Middle and lower dial assemblies.

4.4.7.1 Hammerspring deflection.-The required load shall be applied at the specified location parallel to the vertical axis of the assembly as specified on Drawings D8594468 and D8594469. The hammerspring must return to surface 'A' when released.

4.4.8 Movement subassembly.

4.4.8.1 Kinetic energy of the firing pin.-The subassembly shall be placed in a fixture and tested for kinetic energy by releasing the firing pin from the cocked position. At the conclusion of the test, each subassembly shall be recocked.

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4.4.8.2 Dynamic regulation.-The following procedure shall be observed: The movement subassembly shall be regulated, using the operational procedures furnished with the equipment. The movement subassembly shall be rotated in a clockwise direction when viewed from the bottom at a constant speed of 6000 ± 15 rpm. Timekeeping and performance shall be recorded over a period of not less than 95 seconds. No adjustments shall be made on the escapement during the 95 second check run.

4.4.8.3 Spin function.-The completely wound subassembly shall be spun in a clockwise direction when viewed from number one plate end at a constant rpm as specified on Drawing F8593032. The duration of spin shall be for a period of not less than ten (10) seconds. The movement subassembly shall begin functioning immediately at the specified rpm and the lever assembly shall be released (contacting the periphery of the collar) after spinning for ten (10) seconds. At the conclusion of the test, the subassembly shall be rewound. The function test shall be performed prior to the nonfunction test.

4.4.8.4 Spin nonfunction.-The completely wound subassembly shall be spun in a clockwise direction when viewed from number one plate end at a constant rpm as specified on Drawing F8593032. The duration of spin shall be for a period of not less than three (3) seconds. Nonfunction shall be determined by use of a strobe light. This test shall be the last spin test to which the movement is subjected.

4.4.9 Timing disc torque.-The torque test shall be performed after assembly of the timing disc assembly to the movement and assembly of the locknut on the arbor. The highest and lowest readings will be taken in one uninterrupted complete revolution of the timing disc for determination of compliance with the requirement of Drawing F8594465.

4.4.10 Delay arming mechanism.

4.4.10.1 Spin arming.-The fully wound delay mechanism shall be spun in a clockwise direction when viewed from number one plate end at a constant rpm, for each spin as specified on Drawing F8594387. The duration of spin for each test shall be for a period of not more than fifteen (15) seconds. The time required to arm the assembly shall be recorded within plus or minus 0.1 second. Prior to and at the conculsion of each test, the assembly shall be fully wound and reset. These tests shall be performed prior to the nonarming test.

4.4.10.2 Spin nonarming.-The fully wound delay mechanism shall be spun in a clockwise direction when viewed from number one plate end at a constant rpm as specified on Drawing F8594387. The duration of spin shall be for a period of not less than three (3) seconds. Nonarming shall be determined by use of a strobe light. This test shall be the last spin test to which the delay mechanism is subjected.

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4.4.11 Timer No. 2 and No. 3 assemblies.

4.4.11.1 Setting torque.-Utilizing a ratchet type wrench the setting dial shall be rotated in a clockwise direction when viewed from the top. The highest and lowest readings will be taken on each of two (2) complete uninterrupted revolutions of the dial assembly for determination of compliance with the requirement.

4.4.12 Environmental test.

4.4.12.1 Jolt.-The fuze shall be jolted in accordance with the procedure in MIL-STD-331, Test No. 101.1, except that the fuze shall not be jolted in the horizontal orientation. Nose-up (booster down) shall be the last orientation. Two (2) arms loaded with three (3) fuzes and three (3) dummies shall be used. The fuze shall be disassembled and inspected as follows:

a. Remove peep hole screw. Check engagement of setting pin.
b. Remove booster. Check position of detents and rotor.
c. Remove assembly bolts. Remove movement and DAM assembly. Check security of timing disc retainer, timing disc spring collapse, safety disc relative to firing arm, position of firing pin, and pallet lock slides.
d. Remove DAM. Check position of rotor and condition of detonator.
e. Disassemble major subassemblies. Check all critical parts; i.e., gears, pinions, etc., for broken or deformed pivots, teeth, etc.

4.4.12.2 Jumble.-The fuze shall be jumbled in accordance with the procedure in MIL-STD-331, Test No. 102.1. The disassembly and inspection procedure shall be as outlined in 4.4.12.1.

4.4.12.3 Forty-foot drop.-The fuze shall be dropped (in the metal container) in accordance with the procedure in MIL-STD-331, Test No. 103. The fuze shall be disassembled and inspected as follows:

a. Remove booster. Check position of detents and rotor. Check condition of relay.

4.4.12.4 Transportation-Vibration.-The fuze shall be vibrated in accordance with the procedure in MIL-STD-331, Test No. 104 and as follows:

a. Parallel to longitudinal axis, nose up.
b. First transverse axis to be parallel to a line running through the 100 second marking.
c. Second transverse axis to be parallel to a line running through the 70 second marking.

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4.4.12.4 (Cont'd)

The fuzes shall be disassembled and inspected as follows:

(1) Remove peep hole screw. Check engagement of setting pin.
(2) Remove booster. Check position of detents and rotor.
(3) Remove assembly bolts. Remove movement and DAM assembly. Check security of timing disc retainer, safety disc relative to firing arm, position of firing pin and pallet lock slides.
(4) Remove DAM. Check position of rotor and condition of detonator.
(5) Subject DAM to non-arming test in 4.4.10.2 then arming test in 4.4.10.1.
(6) Subject movement to nonfunction test in 4.4.8.4 then function test in 4.4.8.3.
(7) Disassemble major subassemblies. Check all critical parts, i.e., gears, pinions, etc., for broken or deformed pivots, teeth, etc.

4.4.13 Ballistic tests.-The sample fuzes shall be disassembled and the timer assemblies assembled as test fuzes in accordance with Drawing F10535730. The test fuzes shall be assembled to test rounds (Drawing 10523270) and fired as outlined in 3.6 under the conditions of 4.2.6 and 4.3.3.13, as appropriate (see 6.3.4). Test rounds conditioned at low and high temperature shall be fired within five (5) minutes after removal from the conditioning chamber (see 6.5).

4.4.13.1 Time-to-burst.-Time-to-burst shall be measured to the nearest millisecond using two independent fuze chronograph systems. Two electric clocks or stop watches shall also be used to measure time-to-burst. All measurements obtained in the low and high temperature phases shall be used for information purposes.

4.4.13.1.1 Lot percent defective.-The time values obtained by the chronograph systems shall be used to calculate the estimate of lot percent defective, where both chronograph systems provide a time value, the average shall be taken as the individual round value. Individual round values shall be rounded off to the second decimal place using the method prescribed by the American Society for testing materials. The estimate of lot percent defective shall be calculated using the method prescribed in MIL-STD-414, Example B3. For sample sizes 16 through 18 the estimate of lot percent defective value shall be determined by interpolation between the values given in Table B5, MIL-STD-414 for sample sizes 15 and 20. All appropriate time values obtained for each phase shall be used in the initial calculation of the sample average ($\bar{x}$) and standard deviation (s). The standard deviation value obtained shall then be used to determine statistical outliers (non-normal values) by using the criteria $\bar{x}$ plus or minus 2.88 times the standard deviation. A value judged to be an outlier shall be excluded. The mean and standard deviation shall be recalculated using the remaining values. A minimum of 15 statistically normal values are required for each ambient time-to-burst phase (see 6.3.6).

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4.4.14 Container, air leakage.-The equipment and procedure shall be in accordance with IEL-10535728.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.-Packaging and packing shall be in accordance with Drawings F8794406 and C8794405, unless otherwise directed by the contracting officer.

5.2 Marking.

5.2.1 Metal container.-The metal container shall be marked in accordance with the specific instructions denoted on Drawing F8794406.

5.3 The ICC marking shall be "Percussion Fuzes - Handle Carefully".

6. DEFINITIONS AND NOTES

6.1 Dimensions.

6.1.1 Compliance with tests (proving ground, ballistic performance, and shop) required by the specifications and drawings is mandatory. The contractor will not assume, nor does the government guarantee, that all possible combinations permitted by the tolerance limits of the specifications and drawings will consistently satisfy the test requirements. Therefore, the manufacturer is obligated to choose those combinations of tolerances and fits within the limits of the specifications and drawings that best suit his process needs and still satisfy the testing requirements.

6.1.2 Dimensions which are specifically referred to in the classification of defects, as listed in Section 4, are mandatory. Pertaining to other dimensions, it is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements, but which may provide adequate fuze performance. Therefore, the contractor, at his own risk, may utilize these parts provided there is no adverse effect on fuze functionability. This relaxation of requirements applies to "after the fact" deviations only, to dimensions only, and does not constitute authority to establish new dimensions or to relax efforts to achieve the required sizes within the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsibility in connection with performance of the fuze.

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6.1.3 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

6.2 Armed/unarmed.

6.2.1 Armed.-Rotor locked (spring-loaded) in a position in which the detonator is entirely exposed in the hole into which the firing pin enters.

6.2.2 Unarmed.-Rotor locked in the position in which the detonator is not at all exposed in the hole into which the firing pin enters.

6.3 Ballistic parameters.

6.3.1 Premature burst.-Shall be defined as the occurrence of any of the following:

a. The round detonates in the bore of the gun.
b. The round detonates before 7.0 seconds has expired.
c. The round detonates before 75% of set time.

6.3.2 Outlier-offtime.-An outlier-offtime is a statistical outlier (non-normal value) which falls beyond the appropriate time limits specified in Table I (except when determined to be a premature burst).

6.3.3 Dud.-A dud is considered to have occurred if the test round fails to detonate during flight.

6.3.4 Ballistic test fuze.-A test fuze consists of one timer assembly with booster arranged with two dummy complements, with the original orientation of the timer assembly maintained (see Drawing F10535730), and contained for firing in the test shell, Drawing 10523270.

6.3.5 Electric clock or stop watch values.-Where only electric clock or stop watch time values are obtained and these values indicate a premature function or outlier-offtime, the disposition of the lot shall be referred to the responsible procuring activity.

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6.3.6 Where the number of statistically normal time value obtained in either phase is less than 15 and the lot is not rejected for other reasons, the disposition of the lot shall be referred to the responsible procuring activity.

6.4 Defective Test Samples.-Where sample test fuzes are found to be defective as received at the ballistic facility, all tests of the lot sample shall be held in abeyance pending investigation by the responsible procuring activity.

6.5 Temperature stabilization of fuzes may be checked by a thermal measuring device in good thermal contact with the largest centrally located internal mass. A simulated fuze may be used for this purpose.

6.6 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6.7 Screening of lots.-Caution shall be exercised so no damage occurs to components during static load and torque testing.

6.8 Order data.-Procurement documents should specify the following:

a. Title, Number and Date of this Specification.
b. Level of packing required (see 5).

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